

PROBLEMS SUBMIT STATUS STANDINGS CUSTOM TEST

D. Vasiliy's Multiset

time limit per test: 4 seconds
memory limit per test: 256 megabytes

Author has gone out of the stories about Vasiliy, so here is just a formal task description.

You are given q queries and a multiset A , initially containing only integer 0. There are three types of queries:

- " $+$ x " — add integer x to multiset A .
- " $-$ x " — erase one occurrence of integer x from multiset A . It's guaranteed that at least one x is present in the multiset A before this query.
- " $?$ x " — you are given integer x and need to compute the value $\max_{y \in A}(x \oplus y)$, i.e. the maximum value of bitwise exclusive OR (also know as XOR) of integer x and some integer y from the multiset A .

Multiset is a set, where equal elements are allowed.

Input

The first line of the input contains a single integer q ($1 \leq q \leq 200\,000$) — the number of queries Vasiliy has to perform.

Each of the following q lines of the input contains one of three characters '+', '-' or '?' and an integer x_i ($1 \leq x_i \leq 10^9$). It's guaranteed that there is at least one query of the third type.

Note, that the integer 0 will always be present in the set A .

Output

For each query of the type '?' print one integer — the maximum value of bitwise exclusive OR (XOR) of integer x_i and some integer from the multiset A .

Example

input	Copy
10 + 8 + 9 + 11 + 6 + 1 ? 3 - 8 ? 3 ? 8 ? 11	
output	Copy
11 10 14 13	

Note

After first five operations multiset A contains integers 0, 8, 9, 11, 6 and 1.

The answer for the sixth query is integer $11 = 3 \oplus 8$ — maximum among integers $3 \oplus 0 = 3, 3 \oplus 9 = 10, 3 \oplus 11 = 8, 3 \oplus 6 = 5$ and $3 \oplus 1 = 2$.

Codeforces Round 367 (Div. 2)

Finished

Practice



→ Virtual participation

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Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
261252129	May/16/2024 23:44	Accepted

→ Problem tags

binary search bitmasks data structures trees *1800

No tag edit access

→ Contest materials

- Announcement (en)
- Tutorial (en)

